安装许可证

ONTAP Systems Switches

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安装 BES-53248 集群交换机的许可证

BES-53248 集群交换机基本型号已获得 16 个 10GbE 或 25GbE 端口和 2 个 100GbE 端口的许可。可以通过购买更多许可证来添加新端口。

以下许可证可用于 BES-53248 集群交换机：

许可证类型	许可证详细信息
支持的固件版本	SW-BES-53248A1-G1-8P-LIC
Broadcom 8P 10-25 ， 2P40-100 许可证密钥， X190005/R	EFOS 3.4.3.3 及更高版本
SW-BES-53248A1-G1-16P-LIC	Broadcom 16P 10-25 ， 4P40-100 许可证密钥， X190005/R
EFOS 3.4.3.3 及更高版本	SW-BES-53248A1-G1-24P 许可证
Broadcom 24P 10-25 ， 6P40-100 许可证密钥， X190005/R	EFOS 3.4.3.3 及更高版本
SW-BES542400-40-100G-LIC	Broadcom 6 端口 40G100G 许可证密钥， X190005/R
EFOS 3.4.4.6 及更高版本	SW-BES53248-8P-10G25G-LIC
Broadcom 8 端口 10G25G 许可证密钥， X190005/R	EFOS 3.4.4.6 及更高版本
SW-BES53248/16P-1025G-LIC	Broadcom 16 端口 10G25G 许可证密钥， X190005/R
EFOS 3.4.4.6 及更高版本	SW-BES532424-24P-1025G-LIC
Broadcom 24 端口 10G25G 许可证密钥， X190005/R	EFOS 3.4.4.6 及更高版本

步骤

1. 将集群交换机连接到管理网络。
2. 使用 `ping` 命令验证与托管 EFOS ，许可证和 RCF 文件的服务器的连接。

   此示例验证交换机是否已连接到 IP 地址为 172.19.2.1 的服务器：

```
(cs2) # ping 172.19.2.1
Pinging 172.19.2.1 with 0 bytes of data:

Reply From 172.19.2.1: icmp_seq = 0. time= 5910 usec.
```

3. 检查交换机 CS2 上的当前许可证使用情况：

   s如何许可

```
(cs2) # show license
Reboot needed..................................... No
Number of active licenses......................... 0

License Index  License Type     Status
-------------- ---------------- -----------

No license file found.
```

4. 安装许可证文件。以下示例使用 SFTP 将许可证文件复制到密钥索引 1 。

   重复此步骤可加载更多许可证并使用不同的密钥索引编号。

```
(cs2) #copy sftp://root@172.19.2.1/var/lib/tftpboot/license.dat
nvram:license-key 1
Remote Password:********

Mode............................................ SFTP
Set Server IP................................... 172.19.2.1
Path............................................ /var/lib/tftpboot/
Filename........................................ license.dat
Data Type....................................... license


Management access will be blocked for the duration of the transfer
Are you sure you want to start? (y/n) y

File transfer in progress. Management access will be blocked for the
duration of the transfer. Please wait...


License Key transfer operation completed successfully. System reboot is
required.
```

5. 显示所有当前许可证信息，并在重新启动交换机 CS2 之前记下许可证状态：

s如何许可

```
(cs2) # show license

Reboot needed..................................... Yes
Number of active licenses....................... 0


License Index  License Type      Status
-------------- ----------------- --------------------------------
1              Port              License valid but not applied
```

6. 显示所有已许可的端口：

   s如何全部端口 | 排除断开

   只有在重新启动交换机后，才会显示其他许可证文件中的端口。

```
(cs2) # show port all \| exclude Detach

                 Admin     Physical   Physical   Link   Link    LACP
Actor
Intf      Type   Mode      Mode       Status     Status Trap    Mode
Timeout
--------- ------ --------- ---------- ---------- ------ ------- ------
--------
0/1              Disable   Auto                  Down   Enable  Enable
long
0/2              Disable   Auto                  Down   Enable  Enable
long
0/3              Disable   Auto                  Down   Enable  Enable
long
0/4              Disable   Auto                  Down   Enable  Enable
long
0/5              Disable   Auto                  Down   Enable  Enable
long
0/6              Disable   Auto                  Down   Enable  Enable
long
0/7              Disable   Auto                  Down   Enable  Enable
long
0/8              Disable   Auto                  Down   Enable  Enable
long
0/9              Disable   Auto                  Down   Enable  Enable
long
0/10             Disable   Auto                  Down   Enable  Enable
long
0/11             Disable   Auto                  Down   Enable  Enable
long
0/12             Disable   Auto                  Down   Enable  Enable
long
0/13             Disable   Auto                  Down   Enable  Enable
long
0/14             Disable   Auto                  Down   Enable  Enable
long
0/15             Disable   Auto                  Down   Enable  Enable
long
0/16             Disable   Auto                  Down   Enable  Enable
long
0/55             Disable   Auto                  Down   Enable  Enable
long
0/56             Disable   Auto                  Down   Enable  Enable
long
```

7. 重新启动交换机：

   re负载

```
(cs2) # reload

The system has unsaved changes.
Would you like to save them now? (y/n) y

Config file 'startup-config' created successfully .

Configuration Saved!
Are you sure you would like to reset the system? (y/n) y
```

8. 检查新许可证是否处于活动状态，并注意此许可证是否已应用：

   s如何许可

```
(cs2) # show license

Reboot needed..................................... No
Number of installed licenses..................... 1
Total Downlink Ports enabled..................... 16
Total Uplink Ports enabled....................... 8

License Index  License Type              Status
-------------- -------------------------
------------------------------------
1              Port                      License applied
(cs2) #
```

9. 检查所有新端口是否可用：

   s如何全部端口 | 排除断开

```
(cs2) # show port all \| exclude Detach

                  Admin     Physical   Physical   Link   Link    LACP
Actor
Intf      Type   Mode      Mode       Status     Status Trap    Mode
Timeout
--------- ------ --------- ---------- ---------- ------ ------- ------
--------
0/1              Disable    Auto                 Down   Enable  Enable
```

```
long
0/2               Disable    Auto                 Down   Enable  Enable
long
0/3               Disable    Auto                 Down   Enable  Enable
long
0/4               Disable    Auto                 Down   Enable  Enable
long
0/5               Disable    Auto                 Down   Enable  Enable
long
0/6               Disable    Auto                 Down   Enable  Enable
long
0/7               Disable    Auto                 Down   Enable  Enable
long
0/8               Disable    Auto                 Down   Enable  Enable
long
0/9               Disable    Auto                 Down   Enable  Enable
long
0/10              Disable    Auto                 Down   Enable  Enable
long
0/11              Disable    Auto                 Down   Enable  Enable
long
0/12              Disable    Auto                 Down   Enable  Enable
long
0/13              Disable    Auto                 Down   Enable  Enable
long
0/14              Disable    Auto                 Down   Enable  Enable
long
0/15              Disable    Auto                 Down   Enable  Enable
long
0/16              Disable    Auto                 Down   Enable  Enable
long
0/49              Disable   100G Full             Down   Enable  Enable
long
0/50              Disable   100G Full             Down   Enable  Enable
long
0/51              Disable   100G Full             Down   Enable  Enable
long
0/52              Disable   100G Full             Down   Enable  Enable
long
0/53              Disable   100G Full             Down   Enable  Enable
long
0/54              Disable   100G Full             Down   Enable  Enable
long
0/55              Disable   100G Full             Down   Enable  Enable
long
0/56              Disable   100G Full             Down   Enable  Enable
```

```
long

(cs2) #
```

限制和限制— BES-53248 交换机

如果在安装许可证时出现问题，应先运行以下调试命令，然后再运行 `copy` 命令来安装许可证。

要使用的调试命令：

`debug transferdebug license`

```
(cs2) # debug transfer
Debug transfer output is enabled.
(cs2) # debug license
Enabled capability licensing debugging.
```

在启用 `debug transfer` 和 `debug license` 选项的情况下运行 `copy` 命令时，将返回以下日志输出：

```
transfer.c(3083):Transfer process  key or certificate file type = 43
transfer.c(3229):Transfer process  key/certificate cmd = cp
/mnt/download//license.dat.1 /mnt/fastpath/ >/dev/null 2>&1CAPABILITY
LICENSING :
Fri Sep 11 13:41:32 2020: License file with index 1 added.
CAPABILITY LICENSING : Fri Sep 11 13:41:32 2020: Validating hash value
29de5e9a8af3e510f1f16764a13e8273922d3537d3f13c9c3d445c72a180a2e6.
CAPABILITY LICENSING : Fri Sep 11 13:41:32 2020: Parsing JSON buffer {
  "license": {
    "header": {
      "version": "1.0",
      "license-key": "964B-2D37-4E52-BA14",
      "serial-number": "QTFCU38290012",
      "model": "BES-53248"
  },
  "description": "",
  "ports": "0+6"
  }
}.
CAPABILITY LICENSING : Fri Sep 11 13:41:32 2020: License data does not
contain 'features' field.
CAPABILITY LICENSING : Fri Sep 11 13:41:32 2020: Serial number
QTFCU38290012 matched.
CAPABILITY LICENSING : Fri Sep 11 13:41:32 2020: Model BES-53248 matched.
CAPABILITY LICENSING : Fri Sep 11 13:41:32 2020: Feature not found in
license file with index = 1.
CAPABILITY LICENSING : Fri Sep 11 13:41:32 2020: Applying license file 1.
```

在调试输出中检查以下内容：

- 检查序列号是否匹配： `s`序列号 `QTCU38290012` 匹配。
- 检查交换机型号是否匹配： `model BES-53248 matched` 。
- 检查指定的许可证索引是否以前未使用过。如果已使用许可证索引，则返回以下错误： `License file /mnt/download//license.date.1 already exists.`
- 端口许可证不是功能许可证。因此，应遵循以下语句：在索引为 1.` 的许可证文件中未找到 `功能

使用 `copy` 命令将端口许可证备份到服务器：

```
(cs2) # copy nvram:license-key 1
scp://<UserName>@<IP_address>/saved_license_1.dat
```

请参见 "为 BES-53248 集群交换机安装许可证" 有关可用许可证支持的固件版本的详细信息。

如果需要将交换机软件从 3.4.4.6 版降级，则许可证将被删除。这是预期行为。

在还原到软件的旧版本之前，您必须安装相应的旧版许可证。

编辑参考配置文件（RCF）

要激活新许可的端口，您需要编辑最新版本的 RCF 并取消注释适用的端口详细信息。默认许可证会激活端口 0/1 到 0/16 以及从 0/55 到 0/56 ，而新许可的端口将位于端口 0/17 到 0/54 之间，具体取决于可用许可证的类型和数量。

如果您尝试编辑先前安装的 RCF ，则此过程可能会失败，因为 RCF 中的其他区域已有配置，请参见 [Edit a previously installed RCF file]。

有关可在 BES-53248 集群交换机上使用的许可证类型的详细信息，请参见 "为 BES-53248 集群交换机安装许可证"。

例如，要激活 SW-BES54241-40-100G-LIC 许可证，必须取消注释 RCF 中的以下部分：

```
.
.
!
! 2-port or 6-port 40/100GbE node port license block
!
interface 0/49
no shutdown
description "40/100GbE Node Port"
!speed 100G full-duplex
speed 40G full-duplex
service-policy in WRED_100G
spanning-tree edgeport
mtu 9216
switchport mode trunk
datacenter-bridging
priority-flow-control mode on
priority-flow-control priority 5 no-drop
exit
exit
!
interface 0/50
no shutdown
description "40/100GbE Node Port"
!speed 100G full-duplex
speed 40G full-duplex
service-policy in WRED_100G
spanning-tree edgeport
```

```
mtu 9216
switchport mode trunk
datacenter-bridging
priority-flow-control mode on
priority-flow-control priority 5 no-drop
exit
exit
!
interface 0/51
no shutdown
description "40/100GbE Node Port"
speed 100G full-duplex
!speed 40G full-duplex
service-policy in WRED_100G
spanning-tree edgeport
mtu 9216
switchport mode trunk
datacenter-bridging
priority-flow-control mode on
priority-flow-control priority 5 no-drop
exit
exit
!
interface 0/52
no shutdown
description "40/100GbE Node Port"
speed 100G full-duplex
!speed 40G full-duplex
service-policy in WRED_100G
spanning-tree edgeport
mtu 9216
switchport mode trunk
datacenter-bridging
priority-flow-control mode on
priority-flow-control priority 5 no-drop
exit
exit
!
interface 0/53
no shutdown
description "40/100GbE Node Port"
speed 100G full-duplex
!speed 40G full-duplex
service-policy in WRED_100G
spanning-tree edgeport
mtu 9216
```

```
switchport mode trunk
datacenter-bridging
priority-flow-control mode on
priority-flow-control priority 5 no-drop
exit
exit
!
interface 0/54
no shutdown
description "40/100GbE Node Port"
speed 100G full-duplex
!speed 40G full-duplex
service-policy in WRED_100G
spanning-tree edgeport
mtu 9216
switchport mode trunk
datacenter-bridging
priority-flow-control mode on
priority-flow-control priority 5 no-drop
exit
exit
!
.
.
```

> 对于介于 0/49 到 0/54 之间的高速端口，取消对每个端口的注释，但在 RCF 中仅对其中每个端口的一条 * 速度 * 线取消注释，可以：

- 速度 100G 全双工
- 速度 40G 全双工

如示例所示。

对于介于 0/17 到 0/48 之间的低速度端口，激活相应的许可证后，取消对整个 8 端口部分的注释。

编辑先前安装的 RCF 文件

编辑先前安装的 RCF 文件并运行 `sscript apply` 命令后，可能会收到以下错误消息：

```
(CS1) #script apply BES-53248_RCF_v1.6-Cluster-HA.scr
 Are you sure you want to apply the configuration script? (y/n)
```

选择 `y` 后，您将收到以下错误消息：

```
config
 ...
 match cos 5
 Unrecognized command : match cos 5
 Error! in configuration script file at line number 40.
 CLI Command :: match cos 5.
 Aborting script.
```

要避免或解决此问题描述，您可以选择以下选项之一：

- 要避免此错误，您可以使用以下操作步骤：
  a. 再创建一个仅包含新端口配置的 RCF 文件。
  b. 将第二个 RCF 文件复制到交换机。
  c. 将此脚本应用于交换机。

     s记录应用

- 要解决此错误，请参见知识库文章： "error在应用新 RCF 时，配置脚本文件的行号为 XX"

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